

PRATEEK DWIVEDI

प्रतीक द्विवेदी

KD213, Indian Institute of Technology, Kanpur
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EDUCATION

Indian Institute of Technology Kanpur

Doctor of Philosophy

Computer Science & Engineering

Kanpur, India

2018 - Ongoing

KCC Institute of Technology and Management

Bachelor of Technologies

Computer Science & Engineering

Gr. Noida, India

2013 - 2017

RESEARCH INTERESTS

Broadly I am interested in the Theoretical Aspects of Computer Science. Specifically, my interest lies in Algebraic Complexity Theory. Problems in Algorithms and Cryptography intrigues me as well.

PUBLICATIONS

Demystifying the border of depth-3 algebraic circuits.

with Pranjal Dutta and Nitin Saxena.

62nd Annual Symposium on Foundations of Computer Science (FOCS)

Invited in the special issue of SIAM Journal on Computing (SICOMP)

2021

Deterministic identity testing paradigms for bounded top-fanin depth-4 circuits

with Pranjal Dutta and Nitin Saxena.

36th Computational Complexity Conference (CCC)

The full version is currently under review in SIAM Journal on Computing (SICOMP)

2021

ACADEMIC TALKS AND SEMINARS

Treading the Border Complexity and Identity Testing Paradigms.

PhD State of The Art Seminar, IIT Kanpur

Feb 2022

Deterministic identity testing paradigms for bounded top-fanin depth 4 circuits.

Invited Talk, CS Theory Seminar at Georgetown University.

Oct 2021

Deterministic identity testing paradigms for bounded top-fanin depth-4 circuits.

Conference Talk, CCC'21.

July 2021

Information-Theoretic And Algorithmic Thresholds For Group Testing

PhD Comprehensive Evaluation, IIT Kanpur.

June 2020

On Approximative Closure of Algebraic Complexity Classes

SIGTACS, IIT Kanpur.

Sept 2019

PROFESSIONAL CONTRIBUTIONS

Reviewed research papers for conferences and journals: CCC, ITCS, SODA, SIAM Journal on Computing.

Regularly organise talks and seminars in a special interest group at CSE, IIT Kanpur.

Instrumental in the development of upcoming blog on Algebraic Complexity Theory.

CONFERENCES AND WORKSHOP ATTENDED

62nd Annual Symposium on Foundations of Computer Science (FOCS) *Feb 2022*
Denver, Colorado
Attended Virtually

36th Computational Complexity Conference (CCC), 2021. *July 2021*
Toronto, Ontario, Canada
Attended Virtually

48th International Colloquium on Automata, Languages, and Programming (ICALP) *July 2021*
Glasgow, Scotland
Attended Virtually

52nd ACM Symposium on Theory of Computing (STOC) *June 2020*
Chicago, US
Attended Virtually

Workshop on Algebraic Complexity Theory *March 2019*
ICTS, Bengaluru

ACADEMIC ACHIEVEMENTS

Received financial support from Microsoft Research India. *2022*
To attend Academic conference.

Recipient of financial support from IARCS-ACM India. *2022*
Respectfully declined.

Ranked 23 in Joint Entrance Screening Test(JEST). *2018*

Ranked 1179 in Graduate Aptitude Test in Engineering (GATE). *2017*
Among 1,00,000 candidates.

Presented Undergraduate project at University level Science and Technology Conference. *2017*

Academic Excellence Award. *2014*
KCC ITM, Greater Noida

TEACHING EXPERIENCE

Tutor in CSE, IIT Kanpur

- Introduction to Computing *Mar 2022*
- Data Structures and Algorithms *Jul 2019*

Teaching Assistant in CSE, IIT Kanpur

- Computational Complexity *Jul 2021*
- Modern Cryptology *Jan 2021*
- Mathematics for Computer Science *Jul 2020*
- Data Structures and Algorithms *Dec 2018*
- Introduction to Computing *Jul 2018*

Teaching Assistant in Massive Online Courses

- Introduction to Cryptography *Jan 2022*
eMasters, IIT Kanpur
- Randomized methods in Computational Complexity *Jan 2021*
NPTEL
- Arithmetic Circuit Complexity *Jan 2020*
NPTEL
- Modern Algebra *July 2019*
NPTEL

PERSONAL DETAILS

Name	Prateek Dwivedi (प्रतीक द्विवेदी)
Gender and Pronouns	Male and He/Him.
Permanent Address	A106, Bulland Heights, Crossings Republik Ghaziabad (U.P) - 201016 INDIA
Date and place of birth	21 st Feb 1996, Lucknow, India
Nationality	Indian (भारतीय)